

Clinical indications

Melanin-related hyperpigmentation

Actinic lentigines

Acute and prolonged, unprotected sun exposure may induce benign spotted hyperpigmentation known as actinic lentigines. These localized, benign, flat, and uniformly pigmented lesions are darker than ephelides (freckles) and do not darken or increase in number with sun exposure. Their size is typically larger (≥ 1 cm) compared to simple lentigines, which are usually 2–5 mm in diameter.

Differential diagnosis includes: - Lentigo maligna - Pigmented actinic keratosis - Pigmented flat seborrheic keratosis - Lichen planus-like keratosis

Correct diagnosis is crucial before initiating treatment, as differentiation between benign and malignant lesions can be challenging—even for experienced dermatologists. If there is any clinical uncertainty, a biopsy of the suspicious lesion is mandatory to exclude lentigo maligna.

The removal of actinic lentigines (AL) is often sought for cosmetic reasons. Preventive measures include consistent use of high-SPF sunscreens, avoiding sun exposure during peak UV hours, regular skin cleansing, and maintaining a healthy epidermal barrier. These steps may prevent new lesions and promote spontaneous regression of existing ones.

Therapeutic options for actinic lentigines fall into two main categories: 1. **Photochemical treatments:** cryotherapy, laser, intense pulsed light (IPL), and chemical peels

2. **Topical agents:** target melanin production inhibition, block melanocyte stimulation, and reduce melanin transfer to keratinocytes—though potential side effects exist

Evidence shows that **laser treatments are significantly more effective** than topical therapies or liquid nitrogen for clearing actinic lentigines.

Laser and light treatment

Various laser and intense pulsed light systems (IPLS) can be used alone or in combination with other therapies to improve the appearance of AL. Treatment must be individualized based on skin phototype and pigmentary changes due to acute or chronic environmental exposure.

Two major device categories are used: - **Ultrashort nano/picosecond devices** - **Long-pulsed (LP) systems**

Ultrashort Q-switched (QS) lasers (532 nm, 694 nm, 755 nm) selectively target melanosomes within keratinocytes and melanocytes, minimizing collateral thermal damage—consistent with the principle of selective photothermolysis. In darker skin types, fluences should be minimized to reduce the risk of postinflammatory hyperpigmentation (PIH). A safe endpoint is mild, circumscribed epidermal whitening; micropurpura may appear within 1–2 minutes post-treatment.

Long-pulsed systems, including 595 nm pulsed dye laser (PDL), 755 nm lasers, and IPLS, operate on the extended theory of selective photothermolysis, generating micro-amplified photothermal effects in non-pigmented structures.

Recent evidence supports combining ultrashort-pulsed and LP-IPLS in a single session, significantly improving clinical outcomes and highlighting the benefit of multimodal approaches.

Recommendation

Actinic lentigines are an excellent indication for **ultrashort Q-switched nanosecond/picosecond lasers**, especially when treating a limited number of lesions. Typically, **1–2 treatment sessions** are sufficient for complete clearance.

Lentigo simplex and lentiginosis

Lentigines are light to dark brown macules, ranging in size from <1 mm to several centimeters. They commonly appear in childhood but can develop at any age. Mucosal and nail lentigines are more prevalent in individuals with darker skin types. Lentigines may also be inherited and associated with genetic syndromes such as: - **LEOPARD syndrome** - **Peutz-Jeghers syndrome** - **Carney complex**

These lesions frequently occur on sun-exposed areas and often cause cosmetic concern. However, it is essential to **rule out syndromic associations** before initiating treatment.

Treatment options include: - Topical skin-lightening agents - Chemical peels - Cryotherapy - Laser and light-based devices